

***O.N.I. Acts of Omission:* Rethinking the Role of Educational
Video Games in U.S. History and High School Pedagogy**

AP Capstone Research

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Introduction

Often, in traditional classrooms, history is presented as a series of dates, names, and events to remember. Although textbooks do properly explain causes and effects, how a teacher relates these events to the student varies. Poor presentation can make history seem a little more than terms to remember or tests to take. As a result, many students disengage. UNH Professor of Political Science David Moore laments, “perhaps [students] think this way because that is the way it has been taught to them.”¹ History involves facts and dates, Moore agrees, yet what makes them “historic,” or memorable, is their *meaning*.

Education research continues to investigate ways to engage the student. A lingering issue is that traditional study may appear uninviting to prospective learners. For instance, a cornerstone of historiography is the study of primary sources. One can find these sources largely through reading, which Moore describes as “more like bedtime than like a learning experience.”² Presenting history as a narrative may help minimize the subject’s dullness. One medium that encourages reading, listening, physical engagement, and problem-solving is narrative games. In the book *The Playful Citizen* (2019), subjects play a game to explore the nuances of poverty; after playing, one-third of the players express new interest in political-civic engagement.³ Doubtless games can be effective learning tools.⁴

However, the research base for educational games is split. Most empirical studies of games are studies of gamification, such as of points, badges, competition, or a set time limit. Experiments on educational video games, on the other hand, focusing more on learning than competitive scoring, are lacking. Though it is hard to measure the efficacy of educational games,⁵ there is a growing research, and a growing *need* for research, to develop educational games through gamification.⁶

¹ Moore, David W. “Teaching History.” *The Clearing House*, vol. 62, no. 7, 1989, pp. 303–05.

² Moore, 304.

³ Neys, Joyce, and Jeroen Jansz. “Engagement in Play, Engagement in Politics: Playing Political Video Games.” *The Playful Citizen: Civic Engagement in a Mediatized Culture*, edited by René Glas et al., vol. 1, Amsterdam University Press, 2019, pp. 36–55.

⁴ de Freitas, Sara. “Are Games Effective Learning Tools? A Review of Educational Games.” *Journal of Educational Technology & Society*, vol. 21, no. 2, 2018, pp. 74–84.

⁵ de Freitas, 80.

⁶ de Freitas, 79.

However, game design is not trivial. Humanities professor Rubio-Campillo explains educational games are not as compelling to play as commercial, typical games.⁷ He explores the game *Ancestors: Stories of Atapuerca*,⁸ though its research ended before it was brought to any subjects. Successful game design, such as *Tomb Raider* by Core Design,⁹ engages the player at the cost of historical accuracy. Even if commercial games are historically accurate, like Lucas Pope's *Return of the Obra Dinn*,^{10,11} they prioritize compelling mechanics over teaching history. Moreover, commercial games take too much time, with *Obra Dinn* averaging 8 to 12 hours of playtime to complete;¹² and the steep learning curve in commercial games requires a level of technical "gaming literacy" that not all students are aware of.

Based on current literature, I developed *O.N.I. Acts of Omission*, a historical narrative game based on the United States' history from 1890 to 1936,¹³ using the Unity game engine and Aseprite pixel art program. Testing *O.N.I.* on student subjects, this experiment addressed the following gaps: (1) to design an engaging experience that transmits specific, educational contents for the average student body; and (2) to measure how much an impact the medium of narrative games leaves on the learner, compared to traditional instruction.

Literature Review

1. Educational Games

Video game learning has shifted significantly since the early 2000's, when social stigma around violence in games gave rise to the "serious games movement,"¹⁴ distinguishing games for

⁷ Rubio-Campillo, Xavier. "Gameplay as Learning: The Use of Game Design to Explain Human Evolution." *Communicating the Past in the Digital Age: Proceedings of the International Conference on Digital Methods in Teaching and Learning in Archaeology (12th-13th October 2018)*, edited by Sebastian Hageneuer, Ubiquity Press, 2020, pp. 45–58.

⁸ Murphy's Toast Games. *Ancestors: Stories of Atapuerca*. iOS version, 2018.

⁹ Core Design. *Tomb Raider*. Eidos Interactive, 1996.

¹⁰ Pope, Lucas. *Return of the Obra Dinn*. 3909 LLC, 2018.

¹¹ Pope, Lucas. "Lucas Pope Interview | Obra Dinn Sequels & New Game Mars After Midnight," *YouTube*, uploaded by Cressup, 28 May 2024, at 43:40, www.youtube.com/watch?v=a8IMDfnLULI. Pope explains that, for the development of *Return of the Obra Dinn*, he compiled extensive books on 18th century maritime history and ship construction.

¹² "Game length?" *Steam Community*, 25 Oct. 2018, steamcommunity.com/app/653530/discussions/0/1732090362049393731/.

¹³ Duck Tran, *O.N.I. Acts of Omission*, itch.io, 2026.

¹⁴ Blumberg, Fran C., et al. "The Impact of Recreational Video Game Play on Children's and Adolescents' Cognition." *New Directions for Child and Adolescent Development*, vol. 2013, no. 139, Mar. 2013, pp. 41–50.

commercial, entertainment purposes, and games for serious, educational purposes.¹⁵ This movement has pushed for research to test edutainment in other fields, and is especially central to the research on e-learning against traditional learning.¹⁶

Current studies mostly examine principles of gamification, such as points, player personality, cooperation, time-limit, leaderboards, and badges.¹⁷ In an emergency response training experiment comparing traditional and game-based approaches, Knight et al. concludes the latter group to be more significantly accurate.¹⁸ Similarly, game theorist Kam Star suggests a game's success often depends on how it aligns with user personality.¹⁹

A notable case study at the University of Akron, however, requires an entire college history course to play Sid Meier's turn-based strategy game *Civilization IV*.²⁰ In it, players take the role of a civilization leader (e.g. Peter the Great of Russia), aiming to guide their civilization from 4,000 B.C.E. to the twenty-first century C.E. At Akron, students, after playing, held seminars on historical themes: world systems, cultural bias, determinism, gender roles, colonialism, and economics. Although History Chair Department Martin A. Wainwright reports students' positive responses,²¹ Akron's case lacks a proper control group, and is therefore not a proper experiment to test how much understanding students gain.

Meanwhile, in cases where games are engaging, e.g. commercial games, they are not intended for learning. Historical games, like *Tomb Raider* by Core Design or *Total War* by Creative Assembly,²² trade historical accuracy for entertainment. Likewise, games that make an effort to be historically true, like *Return of the Obra Dinn* by Lucas Pope and the *Assassin's Creed* (AC) franchise by Ubisoft, lack empirical studies on how much learners have gained from them. Moreover, these games are long, complex, designed to be repeated, processed, tried at for

¹⁵ de Freitas, 75.

¹⁶ Knight, James F., et al. "Serious Gaming Technology in Major Incident Triage Training: A Pragmatic Controlled Trial." *Resuscitation*, vol. 81, no. 9, Sept. 2010, pp. 1175–1179.

¹⁷ Danelli, Federico. "Implementing Game Design in Gamification." *Gamification in Education and Business*, edited by Torsten Reiners and Lincoln C. Wood, Springer, 2015, pp. 67-79.

¹⁸ Knight et al., 1175.

¹⁹ Star, K. *Gamification, Interdependence, and the Moderating Effect of Personality on Performance*. 2015. Coventry University, PhD dissertation.

²⁰ Firaxis Games. *Sid Meier's Civilization IV*. 2K Games, 2005.

²¹ Wainwright, A. Martin. "Teaching Historical Theory through Video Games." *The History Teacher*, vol. 47, no. 4, 2014, pp. 579–612.

²² Creative Assembly. *Total War: Napoleon*. Sega, 2010.

“10, 20, or even 200 hours.”²³ At that point, one must ask whether the focus is the game or the subject.

Despite these barriers, a well-designed game can bring out a student’s intrinsic desire to learn independently.²⁴ Professor Kurt Squire, best known for his contribution to the field of game-learning, interviewed kids in an afterschool program hosted by Wisconsin-Madison: after playing Sid Meier’s *Civilization* series—featuring scenarios like Egypt, Phoenicia, ancient Rome—all kids had checked out books on their chosen scenarios, outlined strategies, adopted identities in roleplays at camp, practiced in-game diplomacy (a basic form of political realism), could pinpoint ancient civilizations on a map, and regularly used words like monotheism, cathedral, and ancient Persians.²⁵ Squire’s research shows positive effects of games on learning. However, similar to Wainwright’s *Civilization* college course, Squire’s paper compiles qualitative interviews, and does not compare to any control group.

2. Issues of Gamification

Gamification is the use of gamic principles in non-game contexts, like work, health, education, and training. Kahoot, for example, encourages competition through time-restricted thinking and is reported to have increased focus, understanding, and fun.²⁶

However, gamification can boost immediate engagement without necessarily enhancing deeper conceptual learning. Attali and Arieli-Attali, after investigating how points were awarded for speedy, accurate responses in a math test, concludes that although points *do* increase user response speed, motivational benefits are transient, short-lived, and may not translate to long, sustainable outcomes.²⁷

Although these concerns are valid to consider, gamification may act as a necessary hook, just as Squire’s *Civilization* program may hook a kid’s attention. As such, the research should

²³ Squire, Kurt D. “Video Games and Education: Designing Learning Systems for an Interactive Age.” *Educational Technology*, vol. 48, no. 2, 2008, pp. 17–26.

²⁴ Squire, 19.

²⁵ Squire, 18–20.

²⁶ Lohitharajah, Janarthani, and Punithalingam Youhasan. “Utilizing gamification effect through Kahoot in remote teaching of immunology: Medical students' perceptions.” *Journal of Advances in Medical Education & Professionalism*, vol. 10,3 (2022): 156-162.

²⁷ Attali, Yigal, and Meirav Arieli-Attali. “Gamification in Assessment: Do Points Affect Test Performance?” *Computers & Education*, vol. 83, Apr. 2015, pp. 57–63.

move away from the transient elements of gamification and provoke a deeper inquiry with it instead.

3. Game Design Principles

A recurring challenge for educational games is designing a fun experience. Educational games (known as edutainment) typically and falsely inject too much content over fun, stemming from the misconception that, because learning is so boring, games have to “sweeten the pill.”²⁸ When games are boring, they lack effective game design theory that may include but is not limited to, according to Salen and Zimmermann’s *Rules of Play* (2004): immediate user feedback, clear rules, appropriate challenge, and conflict and outcome.²⁹

In basic game development, a good game loop takes precedence over story. A type of commercial game genre, often termed “bureaucracy thrillers” or “document inspection,” is exemplary of a mundane setting with engaging mechanics. Typically, players drag and drop and interact with documents, examine them, and make decisions based on the contents inside of them. These games are often repetitive, detail-oriented, and incredibly tied to narrative consequence and tactile feedback.³⁰

The first game of this sub-genre is Lucas Pope’s *Papers, Please*. The user plays as a border inspection guard and must verify passports of immigrants or citizens, then to either accept or deny them entry based on the rules in the in-game rulebook, which updates every level, becoming more challenging, convoluted, and morally dubious. Every second, players engage with the interface, scan for clues, compare pixelated images in limited time (see Fig. 1). When players deny or accept entry of some characters, in-game narratives and characters react. When players defy the rules in the rule-book, immediate feedback takes the form of a fine ticket, deducting the player’s salary. Per *Rules of Play*’s gamification principles, *Papers, Please* has a successful net profit of roughly \$12 million.³¹

²⁸ Resnick, Mitchel. “Edutainment? No Thanks. I Prefer Playful Learning.” *Associazione Civita Report on Edutainment*, 2004, pp. 1-4.

²⁹ Salen, Katie, and Eric Zimmerman. *Rules of Play: Game Design Fundamentals*. Mit Press, 2004, pp. 1–694.

³⁰ Ramos et al., 136.

³¹ “Papers, Please.” *Steam Revenue Calculator*, 2024, steam-revenue-calculator.com/app/239030/papers-please.



Fig. 1. Gameplay interface of *Papers, Please* (2013).

Players have to navigate a cluttered desk of documents, check their validity, and deny or allow entry of a character into Arstotzka. Source: 3909 LLC.

In a sister genre, *The Westport Independent* (2016) is a game by Double Zero One Zero Ltd.³² It is a newspaper censorship simulator (see Fig. 2), where the player manages a newspaper agency in an increasingly fascist society. The Loyalist Party of Westport, after imposing the Public Culture Bill in an effort to censor the media, forces the players to balance articles to write, factions to please, and politically-affiliated employees to watch over. However, a common criticism of *The Westport Independent* is its lack of consequence. The *Rock Paper Shotgun* magazine reviews the game to be repetitive and monotonous, becoming less involved as gameplay progresses. Often, there are massive, confusing assortments of mail that have no bearing on anything; there is a lack of, regardless of the player's decisions, any in-game response from the government or the rebel faction. The magazine writes there are "no twists or turns at all."³³ Moreover, unlike the fast-paced drag and drop in *Papers, Please*, *Westport Independent* employs long-form reading that may slow play momentum.

³² Double Zero One Zero. *The Westport Independent*. Coffee Stain Studios, 2016.

³³ Walker, John. "The Westport Independent Review." *Rock Paper Shotgun*, 22 Jan. 2016, www.rockpapershotgun.com/the-westport-independent-review.

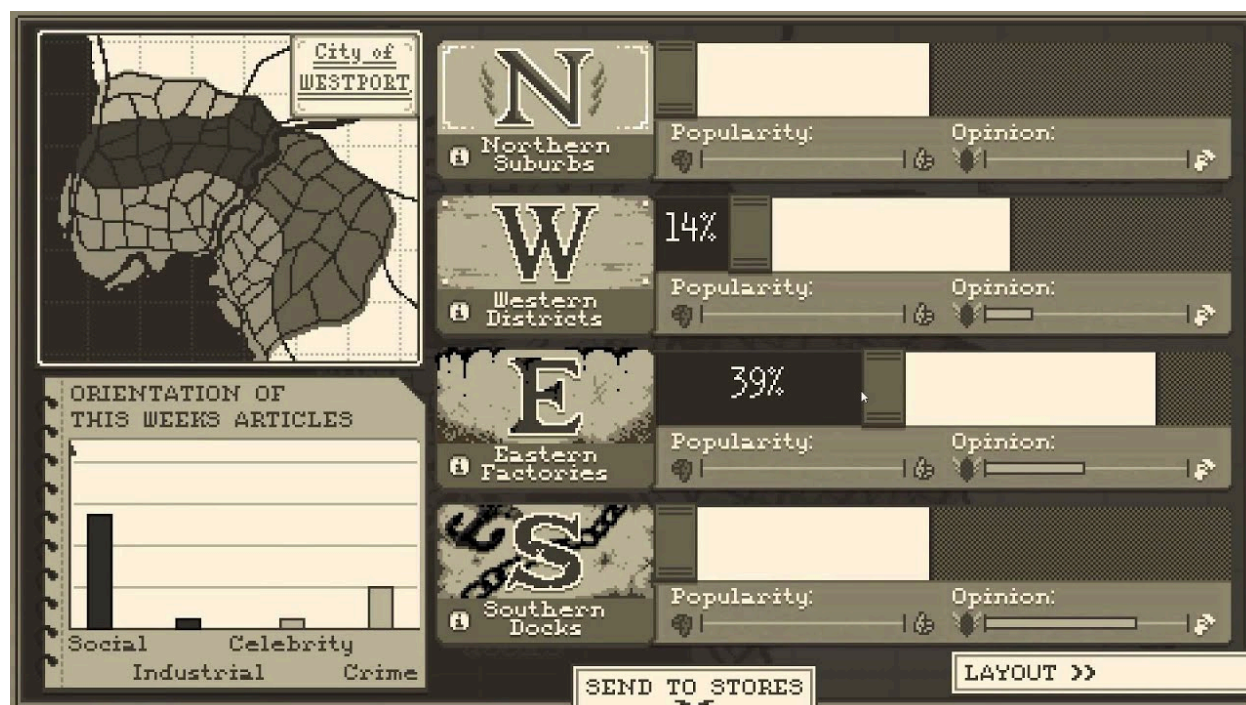


Fig. 2. Gameplay interface of *The Westport Independent* (2016).

The player must appeal to certain districts in the City of Westport, each one with unique traits, needs, and demographics. Source: Double Zero One Zero Ltd.

Successful game designs, therefore, must have a clear and tight feedback loop. Rules must be obvious, so the game, through a tutorial screen in the beginning, should guide the players. The game also must involve tactile, active interaction, such as mouse-clicking, typing, or pressing keys. For an educational simulation to be effective, it must ensure each player interaction is met with appropriate, in-game response.

4. Research Gap

First, much of the foundational research, such as Kurt Squire's scholarship, relies on qualitative data gathered through interviews and anecdotes. Although these studies do demonstrate that games spark an intrinsic will to learn, they often lack a standardized, quantitative bridge to classroom performance. Empirical work also fails to measure results against a control group that uses traditional, didactic instruction, and they do not assess the mastery of specific concepts through a standardized rubric.

Second, current educational games are hard to make and hard to design effectively. Such games prioritize content over a fun experience. Although this project's game is a supplementary tool for didactic instruction, it must still regard gamification principles and user engagement.

This study hopes to address these gaps by developing a historical game simulation and measuring student retention afterwards.

Methodology

This study evaluates whether an interactive simulation designed with immersive mechanics can improve student content retention and engagement more effectively than traditional, didactic instruction. This study expands Rubio-Campillo's research³⁴ on balancing a fun, engaging game with historically accurate contents.

The central research question is: to what extent does a short-form, narrative-driven game, designed with principles of gamification, affect engagement and retention of APUSH high school students at a private high school during the school year 2025–2026?

In order to answer the question, I developed *O.N.I. Acts of Omission* (2026), an interactive simulation designed to teach the complex, historical narratives of foreign relations from 1890 to 1936 in U.S. history. Decisions on game design were made based on Salen and Zimmerman's *Rules of Play* (2003) and studies of commercial games.

All game code and assets used in my study are included in a public GitHub repository for future viewing (Appendix B1).

1. Developing the Game

In *O.N.I. Acts of Omission*, players are 20th-century clerks at the Office of Naval Intelligence (O.N.I.) tasked with transcribing and vetting sensitive documents. By choosing to Send or Discard intelligence, students will trigger a reactive voicemail from a superior officer that reveals the immediate alternative historical consequences of the player's choice.

³⁴ Rubio-Campillo, 47–48.

The game was built with the Unity game engine (v. 2022.3.16f1). All assets were digitally hand-drawn using the Aseprite pixel art program (Appendix D). All sound effects were remixed from Creative Commons-licensed libraries.

a. Mechanics Design

Referencing *Papers, Please* by Lucas Pope, *O.N.I. Acts of Omission* encourages player interactivity by means of typing (Appendix C4), dragging and dropping documents (Appendix C3), and clicking objects on the screen (Appendix C5).³⁵ By typing out each document, players are forced to read and parse primary sources.

Whether the player Sends or Discards a document, he or she will trigger a unique voicemail from a superior officer (See Fig. 4). This mechanic answers the principle of immediate feedback.³⁶ The unique voicemail answers the principle of choices and consequence, establishing meaning for each player decision (Appendix C6).³⁷

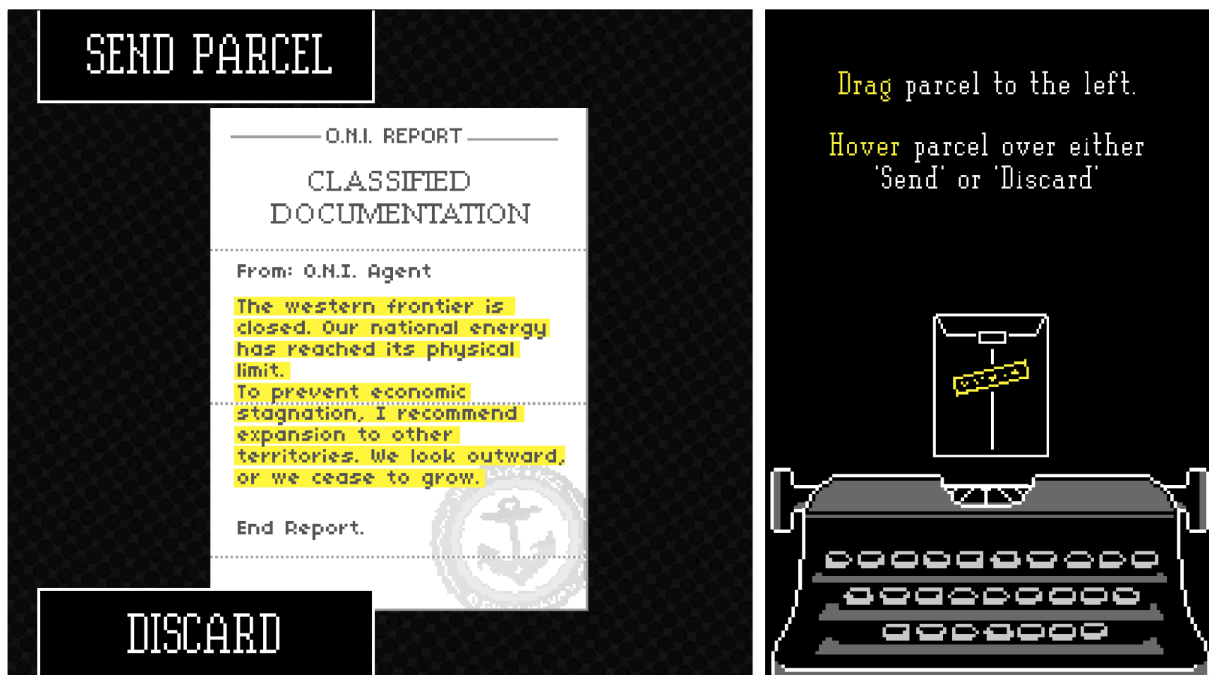


Fig. 3. Gameplay screen of *O.N.I. Acts of Omission* (2026).

³⁵ Salen and Zimmerman, 58–69.

³⁶ Salen and Zimmerman, 87.

³⁷ Salen and Zimmerman, 367.

Player has the choice to either Send or Discard the document after transcribing it.

Source: Author's personal collection.

Each document has a time limit (Appendix C2). The player must transcribe it within the allotted limit. Moreover, because each student has a different typing speed, the time limit adjusts to the WPM calculated by the game on the first document transcribed. This answers the principle of appropriate challenge.³⁸

b. Art Design

O.N.I. Acts of Omission utilizes a minimalist, three-color pixel art aesthetic to evoke an austere 20th-century bureaucratic interface, minimizing visual distractions, inspired by Lucas Pope's black-and-white design for *The Republia Times* (2012)³⁹ (See Fig. 5–6) and ALJO Games's black-white-and-red design for *MORSE* (2025)⁴⁰ (See Fig. 7).

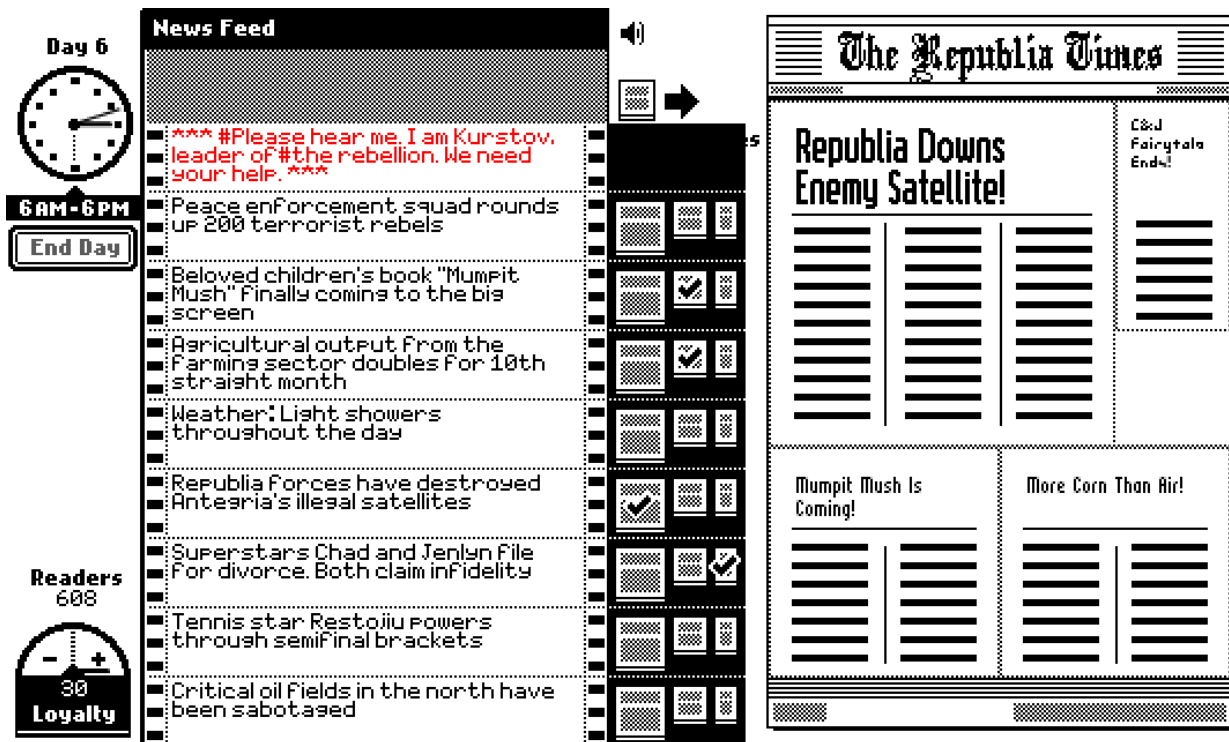


Fig. 4. Gameplay screen of *The Republia Times* (2012).

³⁸ Salen and Zimmerman, 351.

³⁹ Pope, Lucas. *The Republia Times*. Self-published, 2012.

⁴⁰ Johansson, Alex. *MORSE*. ALJO Games, 2025.

Black and white are the dominant colors, with the occasional reds to signify important notices.

Source: Lucas Pope.

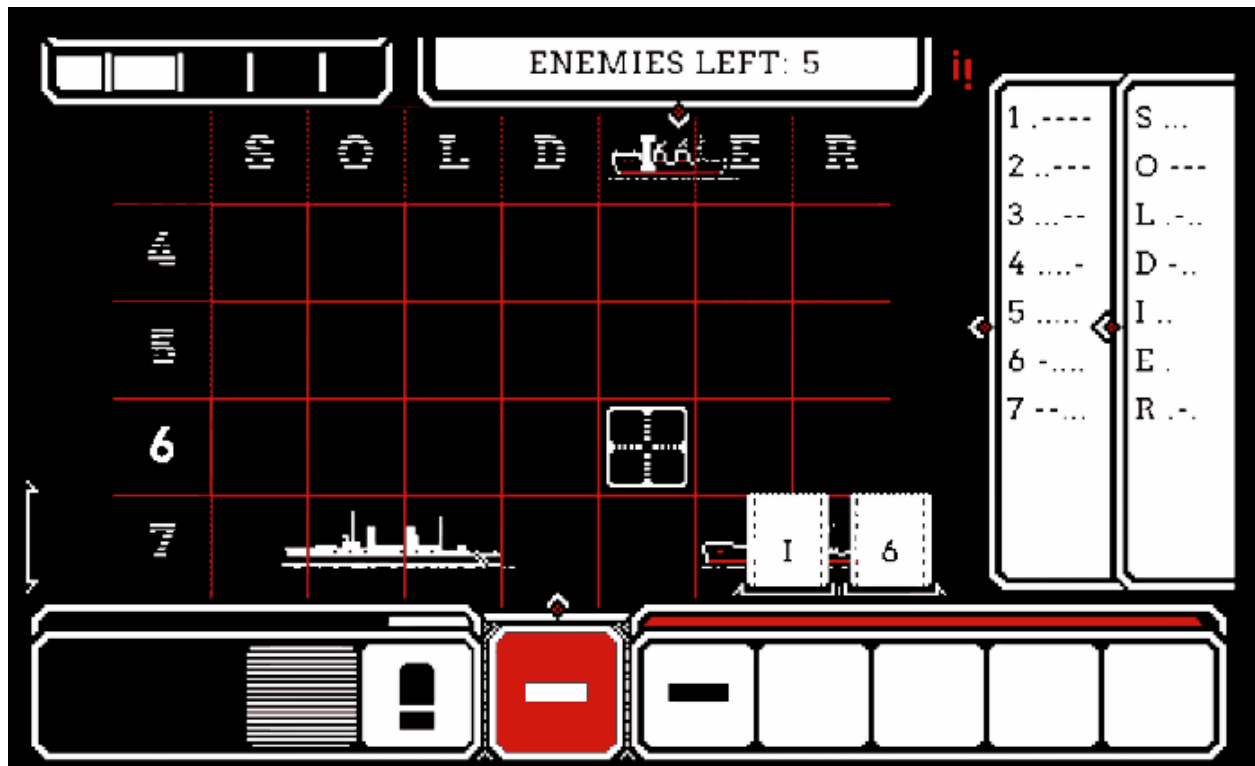


Fig. 5. Gameplay of *MORSE* (2025). Source: ALJO Games.

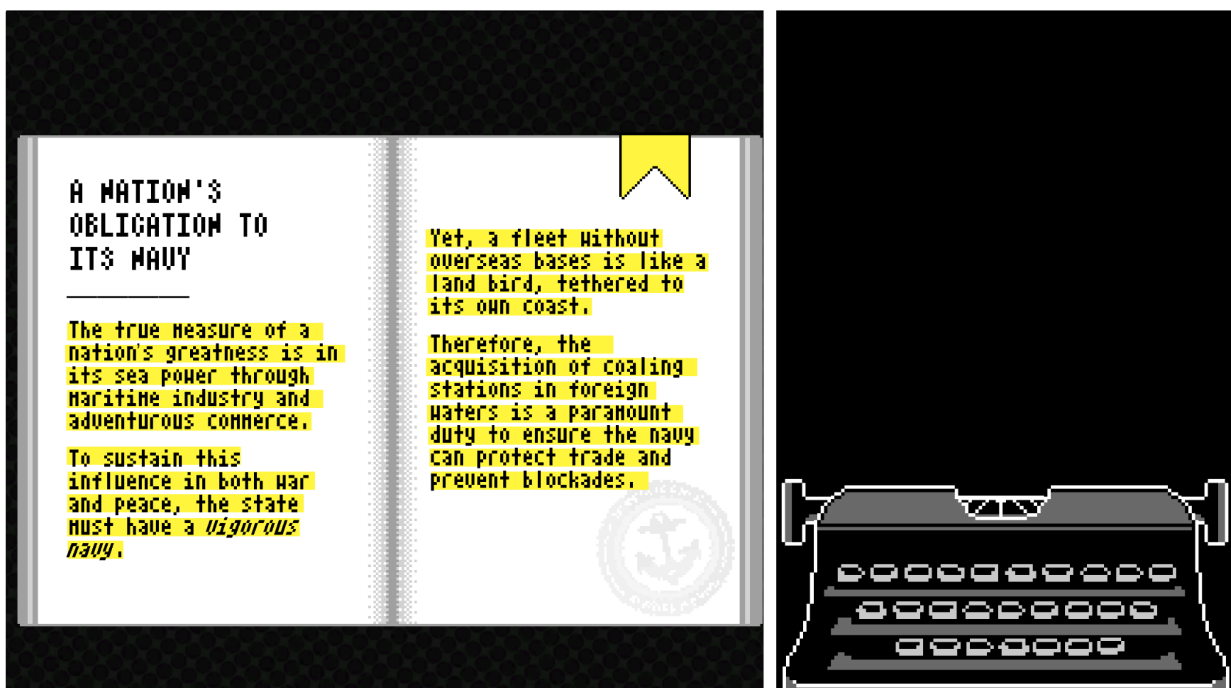


Fig. 6. Gameplay of *O.N.I. Acts of Omission* (2026).

Yellow is used to highlight important texts. Source: Author's personal collection.

c. Gathering History Contents & Resources

To ensure historical veracity, the game's contents were written based on *The American Spirit: U.S. History as Seen by Contemporaries, Volume II* (1998),⁴¹ a textbook that collects primary sources from 1865 onwards and is often used in U.S. history college curriculums. The game specifically addresses U.S. foreign policy from the Spanish-American War to post-WWI, as mapped in the following table:

Table 1. *Mapping Primary Sources to Ingame Contents and Goals*

Primary Source Document	Historical Context	Learning Objective
Turner, Frederick Jackson. <i>The Significance of the Frontier in American History</i> (1893). ⁴²	The western frontier closed, necessitating overseas expansion to prevent economic stagnation.	Reflect on the anti- and pro-imperialism debate of the 1890s.
Pulitzer, Joseph. "The War Must Be Ended." <i>New York World</i> (1897). ⁴³	Yellow journalism incited public outrage and encouraged a war with Spain.	Demonstrate how media manipulation limits diplomatic options.
Norton, Charles E. "Patriotic Protest" (1898). ⁴⁴	The Anti-Imperialist League argued the war on Spain was forced.	Present a moral counter-argument to imperialism.
McKinley, William. "Interview with General Rusling" (1903). ⁴⁵	Annexing the Philippines was framed as a religious and moral duty to civilize the indigenous populace.	Confront the rhetoric of the White Man's Burden.
Anonymous Agent Report (Synthesis of <i>The American</i>	Reports of soldier mutilation and the establishment of	Simulate frontline ethics.

⁴¹ Kennedy, David M., and Thomas A. Bailey, editors. *The American Spirit: United States History as Seen by Contemporaries*. 9th ed., vol. 2, Houghton Mifflin, 1998.

⁴² Turner, Frederick Jackson. *The Significance of the Frontier in American History*. 1893.

⁴³ Pulitzer, Joseph. "The War Must Be Ended." *New York World*, 13 Feb. 1897.

⁴⁴ Norton, Charles E. "Patriotic Protest." *Public Opinion* 24, 23 June 1898, pp. 775–776.

⁴⁵ General James Rusling, "Interview with President William McKinley," *The Christian Advocate* 22 January 1903, 17. Reprinted in Daniel Schirmer and Stephen Rosskamm Shalom, eds., *The Philippines Reader* (Boston: South End Press, 1987), 22–23.

<i>Spirit</i> , pp. 192–93). ⁴⁶	concentration camps during the Philippine-American War.	
“The ‘Oppressed’ of the Philippines.” <i>The Nation</i> (1902). ⁴⁷	An opinion piece argued the U.S. was guilty of the same atrocities for which it once criticized Spain.	Highlight rhetorics often used to justify foreign intervention.
State.gov. “War-Related Debts to the U.S.” (2000). ⁴⁸	Despite neutrality, private American banks loaned billions to the Allies.	Financial entanglements may have roped the U.S. into WWI.
Devine, John. “Historical Congressional Hearings and Reports” (1934–1936). ⁴⁹	Investigations into banks/arms manufacturers influencing U.S. entry into WWI for profit.	Show the weight of financial interests in matters of war.
U.S. Census Bureau. “Historical Census Statistics by Race” (2002). ⁵⁰	African Americans moved from the South to Northern industrial cities during/after WWI.	Show effects of war profiteering to general populace.

There are 10 document assets, each containing a passage paraphrased and shortened from a primary source, as the original text, often spanning pages, is too long for a short-form game (Appendix B1). When paraphrased, the core of the content was kept, while repetitive ideas and repeating, nonrestrictive clauses were removed for brevity, such as the following:

Table 2. *Example of In-game Document Asset, Before and After Trimming Original Content:*

Primary Source	Charles Eliot Norton, "Patriotic Protest," <i>Public Opinion</i> 24, June 23, 1898.
Original Source	And now of a sudden, without cool deliberation, without prudent preparation, the nation is hurried into war, and America . . . compels a weak and unwilling nation to a fight, rejecting without due consideration her [Spain's] earnest and repeated offers to meet every legitimate demand of the United States.

⁴⁶ Kennedy and Bailey, 192–93.

⁴⁷ “The ‘Oppressed’ of the Philippines.” *The Nation*, 1902.

⁴⁸ “Office of Development Finance, Bureau of Economic and Business Affairs: War-Related Debts of Other Countries to the U.S. Government.” *U.S. Department of State Archive*, 1 Mar. 2000, 1997–2001.state.gov/issues/economic/fs_000301_wardebt.html.

⁴⁹ Devine, John. “Research Guides: Historic Congressional Committee Hearings and Reports: Nye Committee (Munitions Industry, 1934–1936).” *Guides.bpl.org*, guides.bpl.org/Congress/Nye.

⁵⁰ Gibson, Campbell, and Kay Jung. *Historical Census Statistics on Population Totals by Race, 1790 to 1990, and by Hispanic Origin, 1790 to 1990, for the United States, Regions, Divisions, and States*. Working Paper no. 56, U.S. Census Bureau, 2002.

Adapted Script	Now suddenly, the nation is hurried into war. America compels a weak nation to fight without consideration of Spain's earnest and repeated offers to meet every legitimate demand of the United States.
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2. Participants & Measures

The study was conducted in Advanced Placement (APUSH) and standard U.S. History classes at a private high school (2025–2026). Group B (Experimental) played the game during a scheduled class period, while Group A (Control) read equivalent primary sources (Appendix A3). To ensure ethical compliance, all participants signed informed consent forms (Appendix A1), and content aligned with APUSH standards.

A quasi-experimental pre-test/post-test design measured comprehension and engagement via two questionnaires (Appendices A2 and A4). Sections 1, 2, and the Addendum were identical across groups. Section 1 established baselines via a 5-point Likert scale on historical interest (to isolate history-hobbyist bias) and five open-ended APUSH Period 7 questions. Section 2 reassessed these five questions post-intervention and included an Addendum testing historical transfer to modern conflicts. Comprehension was scored using a 1–5 APUSH Analytic Rubric. Section 3 measured engagement (Likert scale and short-answer), differing only by referencing either the documents (Control) or the game (Experimental).

Results & Analysis

The study yielded distinct quantitative and qualitative variances between the control group (Group A, $n=6$) and the experimental group (Group B, $n=6$). To address the research questions regarding content retention and student engagement, both descriptive and inferential statistics were applied to the pre- and post-test data.

1. Quantitative Data: Historical Comprehension

a. Historical Comprehension

Historical comprehension was measured by scoring the five open-ended questions on a strict 1–5 APUSH rubric, resulting in a maximum possible score of 25 points per student.

The control group (Group A), which engaged in traditional didactic reading of the primary sources, demonstrated an overall higher baseline of historical knowledge but showed no statistically significant growth after the intervention. Their pre- to post-test scores were:

- Subject A1: 21 → 21 (0)
- Subject B1: 17 → 17 (0)
- Subject C1: 18 → 18 (0)
- Subject D1: 14 → 15 (+1)
- Subject E1: 16 → 15 (−1)
- Subject F1: 12 → 12 (0)

The mean difference (d_1) for Group A is 0.

Conversely, the experimental group (Group B) demonstrated lower baseline scores but measurable post-intervention growth:

- Subject A2: 12 → 17 (+5)
- Subject B2: 9 → 10 (+1)
- Subject C2: 13 → 15 (+2)
- Subject D2: 11 → 13 (+1)
- Subject E2: 12 → 10 (−2)
- Subject F2: 18 → 18 (0)

The mean difference (d_2) for Group B is 1.33.

To determine the statistical significance of Group B's growth, a Paired Samples T-Test was conducted. The calculations are structured as follows:

Standard Deviation of Differences (s):

$$s \approx 2.43$$

T-Statistic (t):

$$t = \frac{d\sqrt{n}}{s} = \frac{1.33\sqrt{6}}{2.34} \approx 1.40$$

Degrees of Freedom (d_f):

$$d_f = n - 1 = 5$$

Critical t -value (at $\alpha = 0.05$):

$$t_{crit} = 2.571$$

Because the calculated t -statistic (1.40) is less than the critical value (2.571), the growth is not statistically significant ($p > 0.05$), likely due to the limited sample size.

However, calculating Cohen's d to measure the effect size yields $d \approx 0.57$. In educational research, an effect size of 0.57 is considered moderate, suggesting the game had a substantive impact on comprehension compared to the control group.

b. Previous Historical Engagement Correlation With Learning

To determine if there is a correlation between a student's baseline enjoyment and engagement of history and how much content they retained from the experiment, results of Likert Scale Question 1–5 (Appendix A2 and A4) were compared to the subjects' total score growth (Post-test score minus Pre-test score).

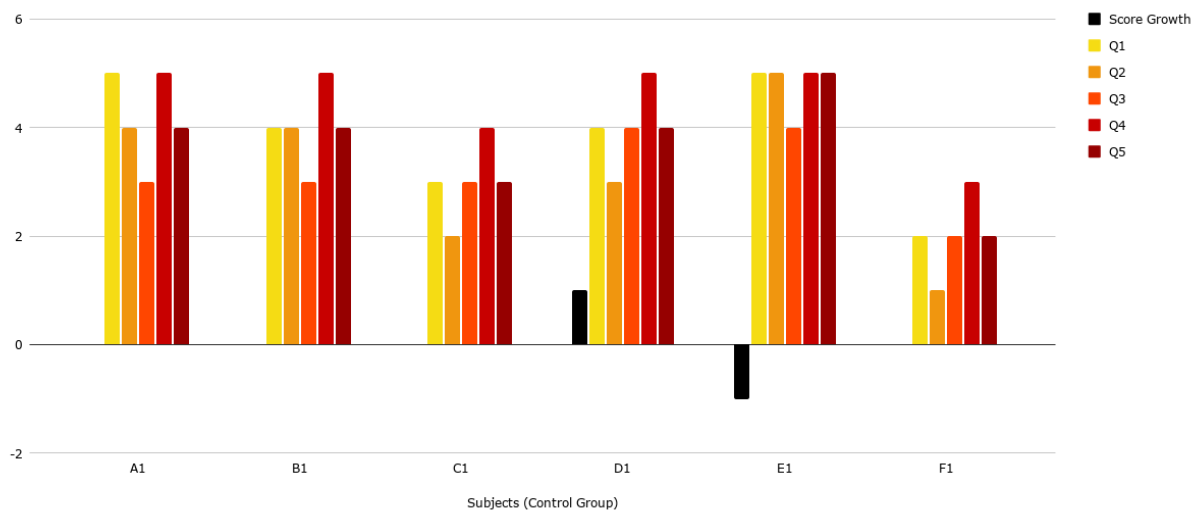


Fig. 7. Correlation Between Pre-Evaluation Interest Profiles and Score Growth in Group A (Control)

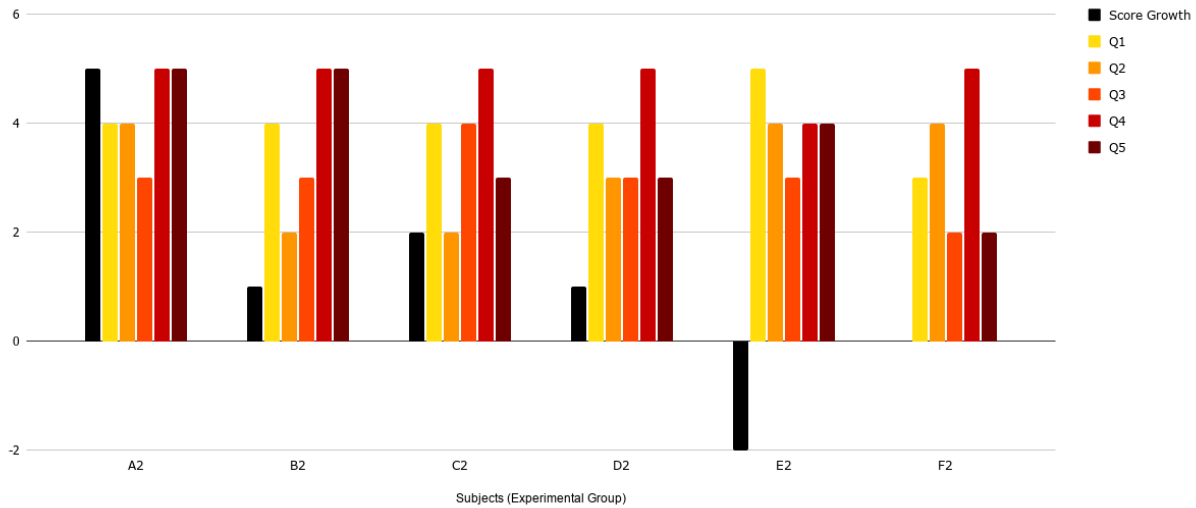


Fig. 8. Correlation Between Pre-Evaluation Interest Profiles and Score Growth in Group B (Experimental)

Q1: General Enjoyment of History

- Group A (Reading): Enjoyment had zero correlation with growth. Subjects A1 and E1 rated their enjoyment a 5, yet scored 0 and -1 growth, respectively.
- Group B (Game): High enjoyment (5) led to a regression (-2). Neutral enjoyment (3) led to stagnation (0). Moderate-to-high enjoyment (4) captured 100% of the positive growth (+1, +2, +2, +5).

Q2: Independent Engagement (Outside of School)

- Group A (Reading): No correlation. Those who engaged independently (4's and 5's) maintained high baseline scores but experienced no post-intervention growth.
- Group B (Game): The two subjects who engage the least outside of school (B2 and C2) both experienced positive growth (+1 and +2). Meanwhile, Subject F2 and E2, who engage heavily outside of school (4's), saw 0 and -2 growth. This implies the gamified mechanics were most effective as a catalyst for students who do not naturally seek out history on their own.

Q3: Specific Interest (Period 7 Early Imperialism)

Both groups primarily answered with 3s. The only experimental subject to rate a 4 (C2) saw a +2 growth, but a control subject who rated a 4 (E1) saw a -1 regression. Specific topical interest did not reliably predict performance in either medium.

Q4: Valuation of History (“Everyone needs to know history”)

Almost all students in both groups rated this a 5 (Strongly Agree). Because there is almost no variance in the data, it cannot be correlated to score growth. The only anomaly is Subject E2, who rated it a 4 and regressed (-2), but this is statistically insignificant.

Q5: Following International and U.S. News

- Group A (Reading): There are no correlations.
- Group B (Game): There is a moderate positive correlation. The two subjects who followed the news the most (A2 and B2, who answered 5) experienced positive growth (+5 and +1). The subject who followed news the least (F2, answering 2) experienced 0 growth. Because the game dealt heavily with journalism, censorship, and political systems, students who already follow modern news likely grasped the game’s mechanics and underlying themes more fluidly.

c. Student Enjoyment of History

To measure students’ interests in history between the experimental group and the control group, three questions were taken from Section 3 of each group’s questionnaire (Fig. 10–11).

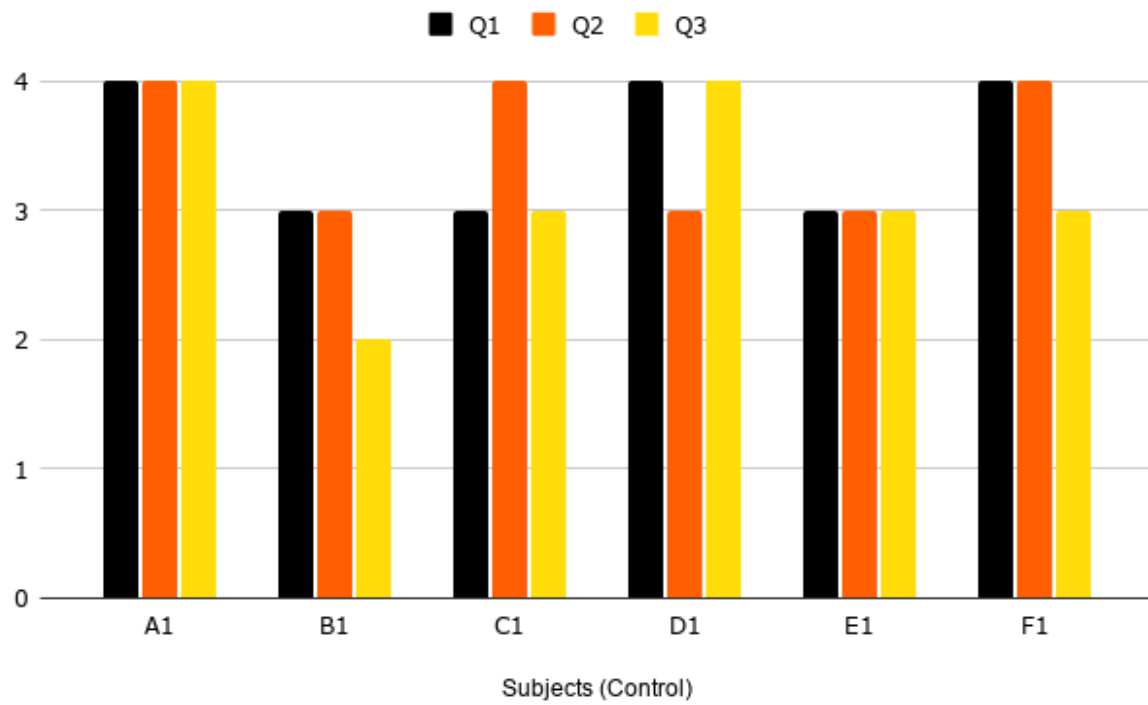


Fig. 9. Learning Engagement Scores for Group A (Control)

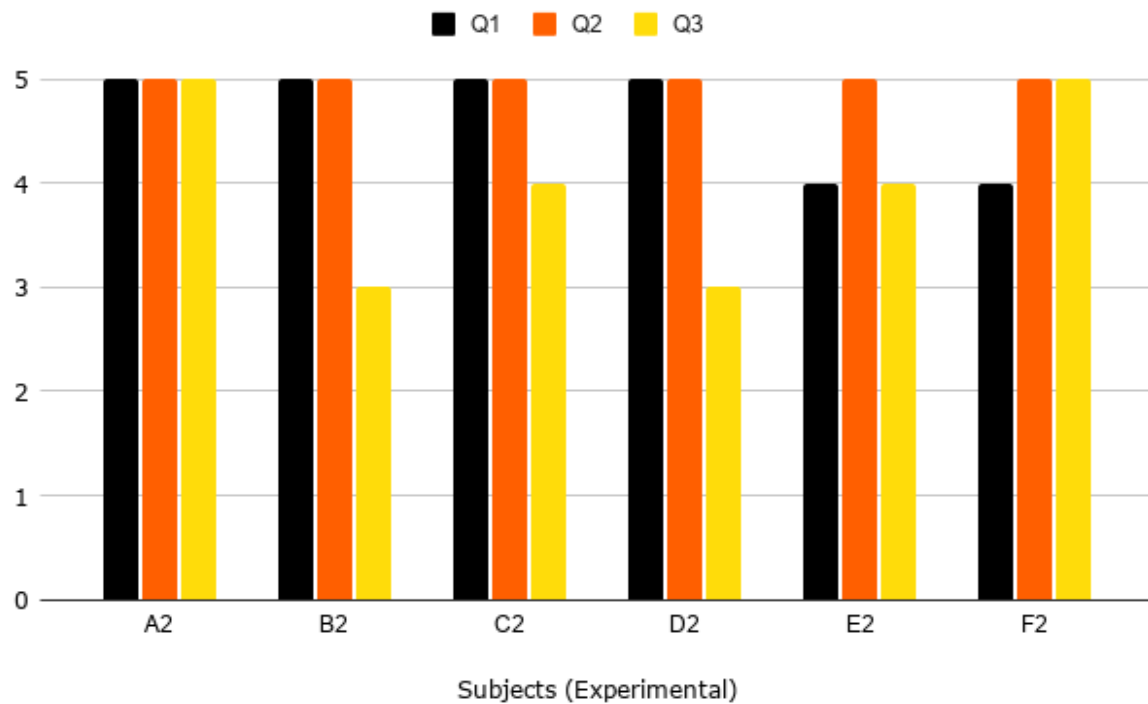


Fig. 10. Learning Engagement Scores for Group B (Experimental)

Comparison A (General Engagement)

- Control Q1: “The provided documents were engaging and made me more interested in U.S. history.”
- Experimental Q1: “The game is engaging and made me more interested in U.S. history.”

Control Mean (μ_C):

$$\mu_C = \frac{4+3+3+4+3+4}{6} = 3.50$$

Experimental Mean (μ_E):

$$\mu_E = \frac{5+5+5+5+4+4}{6} = 4.67$$

The experimental group rated the game a full 1.17 points higher in general engagement compared to the documents. The interactive mechanics successfully elevated the baseline interest of the participants.

Comparison B (Primary Source Interest)

- Control Q2: “Having to read the primary source documents . . . made me more interested in the actual history behind them.”
- Experimental Q2: “Having to read the primary source documents . . . within the game made me more interested in the actual history behind them.”

Control Mean (μ_C):

$$\mu_C = \frac{4+3+4+3+3+4}{6} = 3.50$$

Experimental Mean (μ_E):

$$\mu_E = \frac{5+5+5+5+5+5}{6} = 5.00$$

Because the game places documents inside a narrative where the player must physically interact with them (typing, approving, discarding), every single subject in the experimental group rated their resulting interest in primary sources a perfect 5.

Comparison C (Economic Systems)

- Control Q3: “The documents clearly demonstrated how economic interests . . . directly influence military action.”

- Experimental Q3: “The game clearly demonstrated how economic interests . . . directly influence military action.”

Control Mean (μ_C):

$$\mu_C = \frac{4+2+3+4+3+3}{6} = 3.17$$

Experimental Mean (μ_E):

$$\mu_E = \frac{5+3+4+3+4+5}{6} = 4.00$$

While the game was more effective at demonstrating the link between economics and military action (+0.83 point difference), the experimental group’s scores are slightly lower here than in the other two categories. As observed in the qualitative data (next section), some players became too highly invested in the gameplay, rather than reading the documents.

2. Qualitative Data

a. Engagement and Empathy

In Section 3 of the questionnaire, when asked if the materials were “engaging and made me more interested in U.S. history,” Group B reported a mean score of 4.67/5.0, compared to Group A’s mean of 3.5/5.0.

Short-answer and interview responses revealed Group B experienced high empathy and systemic awareness, frequently abandoning APUSH jargon to focus on human costs. Subject C2 identified in-game consequences of war, stating:

STUDENT C2 But if you had a son go out to war and he died, your life will never be the same. We were sending so many troops, for example, for a city like Montgomery. If you’re a mom and your son went out to war and five of his friends went and five kids your son played sports with growing up. 15 people at war and seven of them die. That's pretty traumatic.

Subject A2 noted the systemic nature of wartime propaganda:

STUDENT A2 Yellow journalism, the game was very clear about . . .
[The government] didn't want to hear about soldiers being

tortured and killed as prisoners of war in these wars . . .
 Always the U.S. had to look big and strong and superior.
 Never any room for fault.

Similarly, Subject B2 noted the game “made me curious on how politicians control the media.”

Subjects also noted the physical mechanics aided focus, with Subject F2 citing an “escape room vibe” and A2 noting the typing challenge forced them to “actually pay attention.”

Interestingly, despite such claims, both Group A and Group B did not extract new, specific information from the documents or from the game. In other words, the learning materials only helped some students recall information already learned in class, rather than helping them connect new themes, despite those themes being present. For instance, a document includes The Nye Committee (1934–1936), which had not been taught in either APUSH or Regular U.S. history. No students referenced the committee or its findings in their post-evaluations. Conversely, when subjects successfully integrated material evidence, such as the Great Migration (Appendix A3), they did so because the topic had been covered in their curriculum.

b. Historical Transfer

The addendum question tests historical application by asking subjects to apply 1890–1930s isolationist and interventionist concepts to modern U.S. tensions with Iran.

Group A yielded heavily polarized results regarding historical transfer. Subject A1 provided an exceptionally detailed analysis and identified the role of modern propaganda and economic drivers (oil and gas prices). Subject A1 also scored 21 on both pre- and post-evaluation. Subject B1 provided a surface-level response, stating “[the U.S.] have different manners to conserve U.S. powers.” Subject B1 scored 17.

All Group B subjects approached the modern conflict with skepticism, and identified the following themes:

- **Media Manipulation:** Subjects who played the game transferred the game’s censorship mechanics directly to their modern political outlook. Subject C2 noted that past wars show how “propaganda and government officials can control people’s point of view” to start worthless conflicts. Subject A2 stated that the public is realizing “they were being

lied to” and that modern intervention is “pointless and it's always going to end the same way.”

- **Economic Drivers:** The game’s focus on trade routes and munitions successfully trained students to look past humanitarian rhetoric. Subject F2 broke down the Iran tension as “adding to our oil supply.” Subject E2 similarly stated the U.S. intervenes purely because it “gets oil and resources.”
- **Pre-Existing Bias:** Interestingly, Subjects B2 and D2 completely ignored the prompt about Iran and went on an extended tangent about the Ukraine-Russia conflict, identifying the military-industrial complex, noting that “[the U.S. are] going to get a lot of money out of a war if people are going to keep on purchasing firearms.” Crucially, when asked if the game helped them reach these conclusions, B2 explicitly stated: “I could have come to it myself just because it’s my background, my viewpoint.”

However, gamification cannot claim superiority over high-level traditional learning. Subject A1’s response proves that a student engaged in traditional reading can achieve an identical, if not superior, level of critical transfer without gamification.

3. Limitations

First, the sample size (n=12) is too small to satisfy strict statistical significance. Second, a 30–50 minute simulation demands a significant portion of a standard class period, posing severe logistical challenges for practical implementation, and this study does not measure long-term retention. Third, the adaptive typing time-limit caused cognitive overload for slower typists (e.g., Subject E2), forcing them to prioritize the physical act of typing over reading comprehension. Consequently, the gamified friction distracted from the retention of specific APUSH jargon.

Conclusion & Discussion

This study set out to determine the extent to which a narrative-driven game *O.N.I. Acts of Omission* could affect engagement and historical retention compared to traditional learning. The findings suggest that while educational games effectively increase student engagement and

systemic thinking, gamification does not overcome traditional learning of jargon and technical knowledge.

Where traditional methods fail, however, is in provoking emotional investment. In *O.N.I.*, players interact with documents and face immediate consequences for their choices, fostering empathy and an applied understanding of how historical systems operate. Players, too, were much more curious about history after the game, confirming de Freitas, Neys, and Jansz's conclusions that games do spark interest in a topic.

Still, because *O.N.I.* applied gamification theories like immediate feedback and challenge, students may have been distracted by gameplay rather than engaging with material. Players did not gain outside-the-classroom knowledge, and the game did not significantly increase retention.

If games cannot teach efficiently and students dislike lectures, what is the next approach? A hypothesis may appear in Subject B2's comment: "I don't like being lectured. I'd rather be more active in learning, to talk it out with the teacher and understand why they did this and this."

This points to a broader shift needed in pedagogy: in a 2026 article by English teacher Jack Liddell, titled "Teaching Shakespearean Sonnets to Brain Rot 15-Year-Olds,"⁵¹ students often disengage *not* because the material is too difficult, but because the traditional format (e.g. listening to a lecture, answering brief questions, and moving on) is entirely passive. In the article, Liddell explains how, when an English teacher presents poems, students approach them as though there are hidden, pre-determined answers, so students very often get stifled. Liddell instead presented a poem he had not read before, chosen by a colleague, and analyzed it alongside his class, discovering students engaged, disagreed, and thought critically of the poems.

The biggest takeaway, according to Liddell, is the lack of critical thinking among students, exacerbated by didactic teaching and reliance on quick, digital software, leading to short-form content and what cultural critic Mo Diggs calls "the dream economy," a forgetful headspace where users rely on immediate gratification.⁵² Liddell advises: "Rather than giving [students] short-form, comfortable content and meeting them where they are (many English teachers now refrain from teaching the full-length novel, opting for excerpts and only short stories), [teachers]

⁵¹ Liddell, Jack. "Teaching Shakespearean Sonnets to Brain Rot 15-Year-Olds." *Substack*, 22 Mar. 2026, <https://jackliddell.substack.com/p/teaching-shakespearean-sonnets-to>.

⁵² Diggs, Mo. "Stop the Stream." *The Metropolitan Review*, 16 Jan. 2026, <https://www.metropolitanreview.org/p/stop-the-stream>.

need to stop somewhere ahead and wait for [students] to catch up.” To Liddell, teachers mistakenly think of fun, creative methods to engage the student, e.g. Kahoot, IXL learning, when in fact “school, in some ways, has to become boring again.”

This presents an interesting area for expansion: comparing traditional, pen-and-paper Socratic learning to modern technology- and software-based methods.

Educational video games should not replace textbooks. Rather, they can set the stage for critical classroom discussions. However, back to Liddell’s point, even though future development in game design should focus on balancing contents and mechanics, pedagogical research should see whether true engagement lies in entertaining students with digital games, or in challenging them to think critically through rigorous, analog mediums, and, more importantly, whether gamified approaches can spearhead further engagement in traditional learning.

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Appendices

Appendix A: Ethical Compliance and Research Instruments

A1. Informed Consent Form

Consent Form

Project Title: *O.N.I. Acts of Omission*: Evaluating the Efficacy of Historical Simulations in U.S. History Education

Researcher: Kim Ha Anh Tran, the Montgomery Academy

Faculty Advisor: Dr. Scott Morris, AP Capstone Research

Introduction

You are invited to participate in a research study involving an educational simulation regarding U.S. History Period 7 (1890–1945). The purpose of this study is to analyze how interactive game mechanics affect student engagement and historical comprehension compared to traditional reading methods.

Procedures

If you agree to participate, you will be asked to:

1. Complete a brief pre-evaluation questionnaire regarding your interest and knowledge of U.S. history.
2. Engage with the assigned learning material.
3. Complete a post-evaluation questionnaire and a brief reflection. The total time commitment is approximately **30-50 minutes**.

Anonymity and Confidentiality

Your participation in this study is **strictly anonymous**.

- **No PII:** No personally identifiable information (such as your name or student ID) will be collected on the questionnaires.
- **Data Coding:** Your responses will be assigned a random numeric code for data analysis purposes only.
- **Storage:** All data will be stored in a password-protected file and will only be accessible to the primary researcher and faculty advisor.

Voluntary Participation and “No-Harm” Clause

Participation is entirely voluntary. You may choose to withdraw from the study at any time without penalty.

- **Non-Punitive:** The results of the questionnaires are for research purposes only. They are **not** a part of your official school grade, and your performance in the simulation or on the history questions will **not be held against you** by your teachers or the school administration.

Risks and Benefits

There are no foreseeable risks involved in this study. Benefits include an opportunity to engage with historical primary sources in a unique format and contributing to the development of better educational tools for future students.

Consent

By signing below, you acknowledge that you have read the above information, are at least 18 years of age (or have provided a signed parental consent form if under 18), and voluntarily agree to participate.

Participant Signature: _____ **Date:** _____

Researcher Signature: _____ **Date:** _____

A2. Control Group Questionnaire: The pre/post evaluations and reading engagement survey.

Questionnaire

Section 1: Pre-Evaluation

Likert Scale (1 = Strongly Disagree → 5 = Strongly Agree)

1. I enjoy learning U.S. history and/or history in general.
2. I engage in learning U.S. History and/or history outside of school.
3. I enjoy learning about Early Imperialism in U.S. history
4. I believe learning history is important, and everyone needs to know history.
5. I follow current international and U.S. news.

Open-ended Questions:

1. In the late 1890s, what was the primary economic motivation for the United States to acquire overseas territories?
2. During the turn of the 20th century, provide one argument used by pro-imperialists to justify expansion and one argument used by anti-imperialists to oppose it.
3. Briefly explain how the media can persuade or dissuade the U.S. government and public to enter the Spanish-American War and the Philippine-American War.
4. Provide one historical example between 1890 and 1930 where the U.S. used a moral or humanitarian justification for a foreign intervention.
5. How did WWI affect the work population of the United States? What socio-economic effects did the Great Depression have on the United States?

Section 2: Post-game Reading Comprehension

Instructions: Answer the following questions again based on your knowledge of U.S. History Period 7 (1890–1945).

1. In the late 1890s, what was the primary economic motivation for the United States to acquire overseas territories?
2. During the turn of the 20th century, provide one argument used by pro-imperialists to justify expansion and one argument used by anti-imperialists to oppose it.
3. Briefly explain how the media can persuade or dissuade the U.S. government and public to enter the Spanish-American War and the Philippine-American War.
4. Provide one historical example between 1890 and 1930 where the U.S. used a moral or humanitarian justification for a foreign intervention.

5. How did WWI affect the work population of the United States? What socio-economic effects did the Great Depression have on the United States?

Addendum: Modern Reflections

The following questions may concern one's political standing. It is acceptable for the questionee to choose not to answer, remain neutral, or not know the answer.

1. Today, the United States is navigating tensions with Iran in the Middle East. Looking back at the intense debates between isolationists and interventionists, how do the arguments for staying neutral versus protecting national interests help explain the difficulties the U.S. faces in deciding whether to intervene in modern foreign conflicts?
-

Section 3: Reading Engagement

Instructions for students: Please rate the following statements on a scale of 1-5 (1 = Strongly Disagree → 5 = Strongly Agree).

Likert Scale Questions:

- The provided documents were engaging and made me more interested in U.S. history.
- Having to read the primary source documents (newspaper clippings, letters, memos) made me more interested in the actual history behind them.
- The documents clearly demonstrated how economic interests (trade routes, banking loans, munitions sales) directly influence military action.

Short Answer Engagement Questions:

- What did you enjoy most about this document analysis/reading experience?
- What would you change or improve about these learning materials?
- How did this specific reading assignment compare to your usual history assignments?
- What did these documents make you curious to learn more about?

And now of a sudden, without cool deliberation, without prudent preparation, the nation is hurried into war, and America . . . compels a weak and unwilling nation to a fight, rejecting without due consideration her [Spain's] earnest and repeated offers to meet every legitimate demand of the United States.

. . . But if a war be undertaken for the most righteous end, before the resources of peace have been tried and proved vain to secure it, that war has no defense. It is a national crime.

Charles Eliot Norton, "Patriotic Protest," *Public Opinion* 24, June 23, 1898.

Document 4

"And one night, late, it came to me this way—I don't know how it was, but it came: (1) that we could not give them back to Spain—that would be cowardly and dishonorable; (2) that we could not turn them over to France or Germany, our commercial rivals in the Orient—that would be bad business and discreditable; (3) that we could not leave them to themselves—they were unfit for self-government, and they would soon have anarchy and misrule over there worse than Spain's was; and (4) that there was nothing left for us to do but to take them all, and to educate the Filipinos, and uplift and civilize and Christianize them, and, by God's grace, do the very best we could by them as our fellowmen, for whom Christ also died. And then I went to bed, and went to sleep, and slept soundly; and the next morning, I sent for the chief engineer of the War Department (our mapmaker), and I told him to put the Philippines on the map of the United States"—pointing to a large map on the wall of his office—"and there they are, and there they will stay while I am President!

General James Rusling, "Interview with President William McKinley," *The Christian Advocate*, January 22, 1903.

Document 5

Even if the condemnation of barbarous warfare in the Philippines by the imperialist press is somewhat belated, we welcome it . . . To fall back on the miserable sophistry that "war is hell" is only another way of making out those who engage in that kind of war to be fiends. It is, besides, to offer an excuse for ourselves which we did not tolerate for an instant in the case of Spanish atrocities.

We made war on Spain four years ago for doing the very things of which we are now guilty ourselves. As the *Chicago News* pointedly observes, we are giving Spain as good reason to

interfere with us on the ground of humanity as we had to interfere with her. Doubtless she would interfere if she were strong enough and thought she could acquire some islands in the virtuous act.

“The ‘Oppressed’ of the Philippines,” *The Nation*, 1902.

Document 6

[President Woodrow Wilson] told [General Huerta] that he must not be a candidate . . . [Wilson] talked about fair elections and constitutional government, and showed a strong desire not only to control Mexican politics but to go into Mexico and regulate the land system of that country . . .

All this for what purpose? Why did we go to Mexico and what did we accomplish and why did we return? We were told that a German vessel was about to land a cargo of guns and ammunition, and this was the reason for hurrying our Navy to Mexican waters. But that same German vessel landed its cargo in Mexico. We are told that our Army and Navy went to Mexico to make Huerta apologize. Has anyone read that apology? . . .

Our policy in Mexico has earned us the contempt of the world, and beyond question has greatly influenced the warring nations of Europe in their present attitude toward us . . .

Congressional Record, 64th Congress, 1st session, January 27, 1916.

Document 7

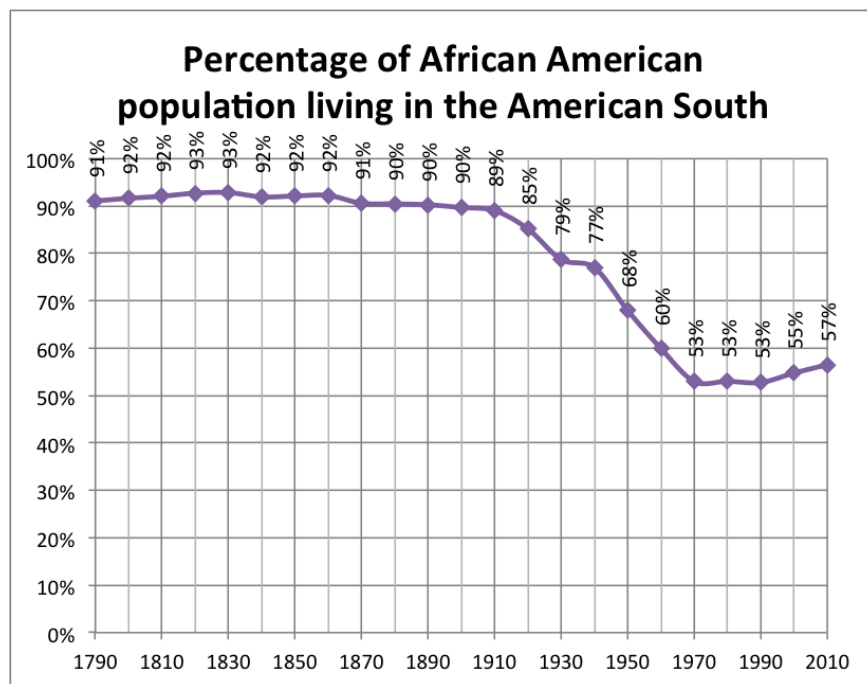
The so-called Senate Munitions Committee came into being because of widespread reports that manufacturers of armaments had unduly influenced the American decision to enter the war in 1917. These weapons suppliers had reaped enormous profits at the cost of more than 53,000 American battle deaths. As local conflicts reignited in Europe through the early 1930s, suggesting the possibility of a second world war, concern spread that these “merchants of death” would again drag the United States into a struggle that was none of its business.

. . . Over the next 18 months, the Nye Committee held 93 hearings, questioning more than 200 witnesses, including J. P. Morgan, Jr., and Pierre du Pont.

. . . Although the Nye Committee failed to achieve its goal of nationalizing the arms industry, it inspired three congressional neutrality acts in the mid-1930s that signaled profound American opposition to overseas involvement.

U.S. Senate Munitions Committee Investigations (Nye Committee), 1934-1936.

Document 8



Campbell Gibson and Kay Jung, *Historical Census Statistics on Population Totals by Race, 1790 to 1990*, U.S. Census Bureau, 2002.

A4. Experimental Group Questionnaire: The pre/post evaluations and the game engagement/mechanics survey.

Questionnaire

Section 1: Pre-Evaluation

Likert Scale (1 = Strongly Disagree → 5 = Strongly Agree)

1. I enjoy learning U.S. history and/or history in general.
2. I engage in learning U.S. History and/or history outside of school.
3. I enjoy learning about Early Imperialism in U.S. history
4. I believe learning history is important, and everyone needs to know history.
5. I follow current international and U.S. news.

Open-ended Questions:

1. In the late 1890s, what was the primary economic motivation for the United States to acquire overseas territories?
2. During the turn of the 20th century, provide one argument used by pro-imperialists to justify expansion and one argument used by anti-imperialists to oppose it.
3. Briefly explain how the media can persuade or dissuade the U.S. government and public to enter the Spanish-American War and the Philippine-American War.
4. Provide one historical example between 1890 and 1930 where the U.S. used a moral or humanitarian justification for a foreign intervention.
5. How did WWI affect the work population of the United States? What socio-economic effects did the Great Depression have on the United States?

Section 2: Post-game Historical Comprehension

Instructions: Answer the following questions again based on your knowledge of U.S. History Period 7 (1890–1945) and the game's contents.

1. In the late 1890s, what was the primary economic motivation for the United States to acquire overseas territories?
2. During the turn of the 20th century, provide one argument used by pro-imperialists to justify expansion and one argument used by anti-imperialists to oppose it.
3. Briefly explain how the media can persuade or dissuade the U.S. government and public to enter the Spanish-American War and the Philippine-American War.
4. Provide one historical example between 1890 and 1930 where the U.S. used a moral or humanitarian justification for a foreign intervention.

5. How did WWI affect the work population of the United States? What socio-economic effects did the Great Depression have on the United States?

Addendum: Modern Reflections

The following questions may concern one's political standing. It is acceptable for the questionee to choose not to answer, remain neutral, or not know the answer.

1. Today, the United States is navigating tensions with Iran in the Middle East. Looking back at the intense debates between isolationists and interventionists, how do the arguments for staying neutral versus protecting national interests help explain the difficulties the U.S. faces in deciding whether to intervene in modern foreign conflicts?
-

Section 3: Game Engagement & Mechanics

Instructions for students: Please rate the following statements on a scale of 1-5 (1 = Strongly Disagree → 5 = Strongly Agree).

Likert Scale Questions:

1. The game is engaging and made me more interested in U.S. history.
2. I understood the historical content better because I could interact with it.
3. The mechanic of choosing to "Send" or "Discard" documents made me think more critically about how historical narratives are shaped by those in power.
4. I would recommend this game to other students studying history.
5. I agree with Pal's character in Chapter 1-2.
6. I agree with Pal's character in Chapter 3-4.
7. Having to read the primary source documents (newspaper clippings, letters, memos) within the game made me more interested in the actual history behind them.
8. The game clearly demonstrated how economic interests (trade routes, banking loans, munitions sales) directly influence military action.

Short Answer Engagement Questions:

1. What did you enjoy most about the simulation experience?
2. What would you change or improve in the game?
3. How did the game compare to traditional learning experiences?
4. What did the game make you curious to learn more about?

Appendix B: Historical Corpus & Narrative Script

B1. *O.N.I.* Game Documentation and Full Script

O.N.I: Acts of Omission (Script)

"Doubtless she would interfere if she were strong enough and thought she could acquire some islands in the virtuous act."

—*The Nation*, 1902.

Core Gameplay Loop (Daily Structure):

1. Player receives a document
2. Player transcribes the document
3. After finishing, player has choice to discard or send the transcription
4. Either option produces voicemail from superior officer that speaks of alternative history

- Green denotes dialogue
- Yellow denotes document names

CHAPTER 1: "The Frontier Closes"

1895, CUBA REBELS AGAINST HER SPANISH COLONIZERS.
 1898, THE *USS MAINE* IS PURPORTEDLY BLOWN BY SPAIN. AMERICA,
 THREATENED, CRIES WAR.

Historical Focus:

- Alfred Thayer Mahan' naval expansion
- Spanish-American War

Pal Script

Good morning, sailor. Congratulations on your new promotion. You are now the clerk for the Office of Naval Intelligence. Your job is simple. We have agents across the country and world. Every day, they bring back news of domestic and international politics. Snippets of newspapers, interviews, magazines.

Your job is to transcribe what these agents send in, so our Admirals, and subsequently our government, can make the necessary decisions.

Once you've completed the report, send the parcel to us. Or discard it, but only if it's detrimental. Good luck.

Documents (3)

1. The Frontier closing [from Agent

From O.N.I. Agent,

The western frontier is closed. Our national energy has reached its physical limit. To prevent economic stagnation, I recommend expansion to other territories. We look outward, or we cease to grow.

End Report.

Turner, Frederick Jackson. *The Significance of the Frontier in American History*. 1893.

- **Send**: Good job. It seems you're getting the hang of work. You know, there is a trend in our history: our forefathers have always been bold and daring adventurers, always expanding to the west. Even if there are obstacles, like Native Americans. Anyway, the Admirals are now looking at your recommendation. Reminds me, you've heard of Admiral Thayer's essay, yes? In case you haven't, he's convinced that our Navy needs to be out there, competing with other European powers. I figure he's right. We need a strong foreign presence, anyway.
Oh, and by the way, new workplace policy: every document is now on a timer. Speed and accuracy are mandatory. Don't let the clock run out. Keep up the work, sailor.
- **Discard**: You didn't think stagnation is worth mentioning? Sailor, let me give you a history lesson: Germany, Britain, and the Netherlands, since the turn of the century, have been taking a piece of Africa and India and Southeast Asia. Meanwhile, where is our fleet? Sailor, you need to focus and send us the parcel. Got it?
Oh, and by the way, new workplace policy: every document is now on a timer. Speed and accuracy are mandatory. Don't let the clock run out. Get working.

2. Newspaper: USS Maine

A picture, sinking of USS Maine. The newspaper is yellow. The headline reads

DESTRUCTION OF WAR SHIP MAINE WAS WORK OF AN ENEMY

Within the same newspaper:

SPAIN GUILTY! MUST DECLARE FOR WAR
HOW LONG ARE THE SPANIARDS TO DRENCH CUBA WITH THE BLOOD AND
TEARS OF HER PEOPLE?
HOW LONG SHALL THE UNITED STATES SIT IDLE AND INDIFFERENT WITHIN
HEARING OF RAPINE AND MURDER?

Pulitzer, Joseph. "The War Must Be Ended." *New York World*, 13 Feb. 1897. *American Documents*,

d1lexza0zk46za.cloudfront.net/history/am-docs/war-must-be-ended.pdf.

Pulitzer, Joseph. "The War Must Be Ended." 1897. *The American Spirit: United States History as Seen by Contemporaries*, edited by Thomas A. Bailey and David M. Kennedy, 9th ed., vol. 2, Houghton Mifflin, 1998, pp. 178-79.

- **Send**: It's ugly, but kicking Spain out of the hemisphere is the quickest way to secure the Caribbean trade routes once and for all. Keep submitting those documents.
- **Discard**: Keeping the *Maine* quiet? Noble, are we? But we need those islands, sailor. News just came from D.C. that our commercial rivals in the Orient is already carving pieces of the Chinese Market. We have American interests there to protect, not to be undercut. Get to work.
- **Late**: Type faster. The *Maine* is at the bottom of the ocean and the Navy is ready to move, with or without your transcript. I've pushed the Guilty verdict through. Wake up.

3. Anti-Imperialist League pamphlet

Now suddenly, the nation is hurried into war. America compels a weak nation to fight without consideration of Spain's earnest, repeated offers to meet every legitimate demand of the United States.

Norton, Charles E. "Patriotic Protest." *Public Opinion* 24, 23 June 1898, pp. 775-776.

Norton, Charles Eliot. "Patriotic Protest." 1898. *The American Spirit: United States History as Seen by Contemporaries*, edited by Thomas A. Bailey and David M. Kennedy, 9th ed., vol. 2, Houghton Mifflin, 1998, pp. 182-83.

- **Send**: Oh, you didn't really need to send that. You've got the youth throwing a fit now, when we need a nation united. You have to understand: we know what we're doing is . . . questionable. But think of the new resources that this can bring to our people. If we get the Philippines, we can trade in China. If we get Puerto Rico, we can get our American influence out there. We need

respect. Better us controlling the seas than the Spaniards or Brits or the Orientals, right?

- **Discard**: Good call dumping that Anti-Imperialist bleeding-heart nonsense. We're mobilizing the Navy tomorrow. Tell you what, sailor, just between us two, we *know* what we're doing is . . . questionable. But *think* of the new resources that this can bring to our people. If we get the Philippines, we can trade in China. If we get Puerto Rico, we can get our American influence out *there*, get our respect out *there*.
- **Late**: Stop debating the ethics of war while the clock ticks. Type faster.

CHAPTER 2: "A Splendid Little War"

SPAIN CEDES CUBA, THE PHILIPPINES, VARIOUS OTHER TERRITORIES TO
AMERICA
1899, U.S. CONTRADICTS PHILLIPINE'S ASPIRATION FOR INDEPENDENCE.
INSURRECTION BREAKS OUT.

Historical Focus:

- Spanish-American War
- Yellow journalism
- Philippine-American War
- Open Door Policy

Pal Dialogue

Welcome back, sailor. Did you read the news?

We've won Cuba and the Philippines from the Spaniards. The papers are calling it a 'splendid little war,' and for once, they're right.

Speaking of which, my own boy just shipped out to Manila this morning. I tell you, seeing him in that uniform . . . it makes a father proud to know his blood is out there bringing civilization to the edge of the map.

However, the job is not done. Filipino General Emilio Aguinaldo is holding an insurrection against us. Don't get bogged down in the 'humanitarian' fluff in those letters from the front; it only slows down the flow of progress. Keep it going, sailor.

Documents (3)

1. China Open Door Policy by Secretary of State John Hay

(Attached with picture of China's port and South East Asia)

To: ONI Field Agents

Re: Chinese Commercial Policy Department

We propose an Open Door Policy in China for the following reasons:

- Preventing serious conflicts of interest among world powers in the Orient, alongside American business and citizen rights in the region.
- No nation should get a head start on treatment of commerce and navigation. The Open Door Policy is therefore equal.
- IF, following success in the Philippines, we must secure our piece of the Chinese market, the most coveted prize in Asia

Hay, John. "The Open Door in China (1899-1900)." *U.S. Department of State, Papers Relating to Foreign Relations of the United States, 1899, 1901*, pp. 129-130. *Washington Unified School District*, rivercity.wusd.k12.ca.us/documents/Academics/Social-Science/GRADE-11-US-HISTORY/PRIMARY-SOURCE-DOCUMENTS/7301864765348308297.pdf.

- **Send**: Secretary Hay is thrilled you fast-tracked his Open Door memo. We just wedged our foot firmly in the Asian market. American businesses there are really thriving, all thanks to you. Keep it up.
- **Discard**: Rule one of economics: if we don't use the Philippines as a stepping stone to secure that Chinese market now, the Brits and Germans are going to wall the whole place off. Look at it this way: they buy the silk and tea, they sell it back to us at five times the price. And our farmers are going to empty, those, pockets. Sailor, why can't you do your job?
- **Late**: The Germans and Brits are already carving up ports of China while you're fumbling with your typewriter. I've sent Hay's memo up. If we miss the Chinese market because you're slow, it's our necks on the line, not just yours.

2. Agent in the Philippines Report

[Anonymous Report from Agent]

Some Filipino tribesmen have horribly mutilated our captured soldiers. Morale is low. We have built concentration camps to keep order among our prisoners.

End Report.

From *The American Spirit*, pg. 192-193.

- **Send:** I told you to keep the fluff out, but you just had to send up the report about us building concentration camps, didn't you? Sailor, have you been . . . drinking on the job? Is your family alright?
- **Discard:** What happens in the jungle stays in the jungle.
- **Late:** This is sensitive material, sailor! You can't just leave reports about concentration camps sitting on your desk for anyone to see. Christ, a whistleblower passed by. Next time, move faster.

3. The Nation Denounces Atrocity

[Map of Philippines and China]

We made war on Spain four years ago for doing the very things of which we are now guilty ourselves. We are giving Spain as good reason to interfere with us on the ground of humanity as we had to interfere with her. Doubtless she would interfere if she were strong enough and thought she could acquire some islands in the virtuous act.

"The 'Oppressed' of the Philippines." *The Nation*, 1902. *The American Spirit: United States History as Seen by Contemporaries*, edited by David M. Kennedy and Thomas A. Bailey, 9th ed., vol. 2, Wadsworth, 1998, p. 193.

- **Send:** Here's some statistics for you: *The Nation* just gained a raving audience after their Philippines article went on the first page. Because of you, anti-imperialists are publicly rebuking us. [Soft laugh] I'm pretty sure anyone bright enough would see that nothing in big politics is in the name of humanity, so what are these people complaining about?
- **Discard:** Right. Into the fire with that one. [soft Laugh]. I'm pretty sure anyone bright enough would see that nothing in big politics is in the name of humanity. But oh well, just toss it in the trash, and the public will never know truth. Except you.
- **Late:** Keep the pace or find a new job.

CHAPTER 3: "Moral Imperialism"

1902, AMERICAN-PHILIPPINE CONFLICT ENDS. TERRITORY ANNEXED.

1912, WOODROW WILSON ELECTED PRESIDENT.

1914, WAR BREAKS OUT IN EUROPE.

Historical Focus:

- Woodrow Wilson's intervention in Huerta as "moral imperialism," declaring it government of "butchers."
- Germany sends Zimmerman telegram to Mexico
- WWI neutrality debate
- Banking loans to Allies
- Lusitania

Pal Dialogue

[Cough] Oh, uh. Good morning, sailor. Sorry, I'm just a bit sick today.

. . . Actually, no. I'm just tired. It's been a long decade since Manila. You know, my boy went over there. To the Philippines. He didn't come back. Died spreading "civilization," you know. Uhm. This is difficult . . .

I don't know, sailor. I'm kind of boxed in.

Ahem. So. Good morning to our new president, Woodrow Wilson. News just arrived from Mexico. General Victoriano Huerta has violently seized power, and Wilson is publicly calling his regime a bunch of butchers. Listen. Fact is, Wilson just doesn't want him in office, but the brass needs to justify our military presence in Mexico to protect American corporate assets. And look out for the war brewing in Europe. There's a lot of talk of intervention.

Handle these files with care, sailor. And look out for signs of war for profit.

Documents (2)

1. Newspaper: "War Erupts in Europe"

WAR ERUPTS IN EUROPE

Wilson's heart goes out to suffering Allies.

The nation retains neutrality.

[Different column]

"NEUTRAL" INTERVENTION?

Major American bank J.P. Morgan has loaned over \$2 billion to the Allies war effort in Europe.

"Office of Development Finance, Bureau of Economic and Business Affairs: War-Related Debts of Other Countries to the U.S. Government."

U.S. Department of State Archive, 1 Mar. 2000,
1997-2001.state.gov/issues/economic/fs_000301_wardebt.html.

- **Send**: Why am I not surprised Mr. Morgan would loan to the Allies? The isolationists are furious about this. They want to investigate the banks now. My pal from Congress is telling me we're pushing for more neutrality. Well, sailor, seems you don't need to pack your bags after all. We might stay home for Christmas.
- **Discard**: Congress remains neutral, but a little birdie from Wall Street is telling me some big names are loaning a bunch of cash to the Brits. The public doesn't know it yet, but this war's going to tug us sooner or later when our money's on the line.
- **Late**: We don't have time for a slow clerk. I've sent the neutrality brief up, but keep in mind: money moves faster than you do.

2. Lusitania sinking & Zimmerman Telegram

BARBARIC GERMAN U-BOAT SANK RMS *LUSITANIA*. 128 AMERICANS DEAD.
The public cries outrage!

IS WAR INEVITABLE?

Intercepted German telegraph reveals a proposal to Mexico:
declare war on the United States, then Germany will help them
reclaim the American Southwest.

- **Send**: This morning going to work, all I saw on the streets were newspapers proclaiming war, war, war. It seems everyone's giving the OK for U.S. involvement. Even my next door grandma. Do they know what they're getting into?
- **Discard**: The Kaiser is still attacking without consequence on international waters, sinking the Brits, the Allies. Even our neutral ships. We might stay out of the trenches today, but we're letting our sailors become target practices. Something tells me we can't stay neutral forever.
- **Late**: 128 Americans are dead and you're frozen? The public is ready to fight, it's a powder keg about to explode. If you're playing the game, you need to keep up with the game.

CHAPTER 4: "Merchants of Death"

1919, TREATY OF VERSAILLES ENDS GREAT WAR. U.S. REJECTS LEAGUE OF NATIONS, RETURNS TO ISOLATIONISM. ROARING 20'S PROCEEDS.

Historical Focus:

- Nye Committee (1934-36)
- Isolationism and Neutrality Acts

Pal Dialogue

Welcome back, sailor. The papers are calling today a happy day. The boys are finally coming home from Europe. Well... the ones who get to come home, anyway.

But behind closed doors, Washington is sweating bullets. Senator Nye is finally digging through the records of every private munitions manufacturer from here to Delaware. It seems the public is just now starting to ask the question you and I have known the answer to since Manila: did these corporations drag our nation into the Great War just to pad their bottom line?

Blood money. That's the thing. So keep your head down, sailor. Read these files carefully. We both know that in this town, the truth isn't what actually happened. It's whatever is in the final transcript.

Documents (2)

1. Nye Committee hearing transcript excerpt

[Report from Senate Munitions Committee Investigations]

After 93 hearings, the Committee has found extensive evidence that the Great War has reaped enormous financial benefits for private Americans ammunition manufacturers, at the cost of over 53,000 American deaths.

[Ubiquitous picture of war profits chart]

While we have uncovered no evidence of a formal conspiracy by these private corporations to trigger our entry into the conflict, it is clear their financial ties to the Allies may now allow war to be a commercial necessity.

See our recommendations:

[Document 2 attached]

"U.S. Senate: 'Merchants of Death.'" [www.senate.gov](http://www.senate.gov/about/powers-procedures/investigations/merchants-of-death.htm),
www.senate.gov/about/powers-procedures/investigations/merchants-of-death.htm.

Devine, John. "Research Guides: Historic Congressional Committee Hearings and Reports: Nye Committee (Munitions Industry, 1934-1936)." *Guides.bpl.org*, guides.bpl.org/Congress/Nye.

Neutrality drafts (attached to document 1)

RECOMMENDATIONS FOR NEUTRALITY

- Renounce Fourteen Points and League of Nations.
- Proposing the Neutrality Acts: mandate embargo on all ammunition and inventory of war, to and from the U.S.
- **Send**: Back in the day, guy who proclaims war at least had the decency to bleed in the mud himself, you know? Anyway, Congress is drafting that embargo. It's going to hurt the market and cost jobs, sure, but if it stops us from acting as Wall Street mercenaries, maybe it's worth the economic hit. At some point, you just have to ask yourself, what's the cost of being top dog in the game?
- **Discard**: I see Senator Nye's investigation quietly disappeared from the morning briefings. Playing it safe, huh? I don't blame you. But, just to tell you what this means: means people aren't looking too closely at the enormous financial benefits those banks raked in. Means, without the Neutrality Acts, we stay tangled up in endless global affairs. At some point, you just have to ask yourself, what's the cost of being top dog in the game?
- **Late**: Senator Nye is on a warpath and you're acting like we have all day. This office is for workers, not statues.

3. The Great Migration chart

Percentage of African Americans living in the South following The Great War.

[Attached chart picture]

https://en.wikipedia.org/wiki/Second_Great_Migration_%28African_American%29#/media/File:Percentage_of_African_American_population_living_in_the_American_South.png

Gibson, Campbell, and Kay Jung. *Historical Census Statistics on Population Totals by Race, 1790 to 1990, and by Hispanic Origin, 1790 to 1990, for the United States, Regions, Divisions, and States*. Working Paper no. 56, U.S. Census Bureau, 2002.

- **Send**: [NONE]
- **Discard**: [NONE]

ENDING

1929, JOB MARKET CRASHES ON "BLACK TUESDAY."

YOU RECEIVE A LETTER. IT IS FROM O.N.I.

Scene change, player receives letter from Pal:

Budget cuts. Your services are no longer required.

-Pal.

Appendix C: Game Code

C1. Open-Source Repositories:

Source Code (GitHub): https://github.com/KimHaAnhTran/ONI_ActsOfOmission.

Playable Web Build (Itch.io): <https://ducktran.itch.io/oni-acts-of-omission>.

C2. Time Limit Manager Code

This script implements the time-pressure variable of the experiment. It forces players to complete transcriptions within a dynamically calculated time limit. The following is a snippet of the script `TimerManager.cs` with its primary functions displayed:

```
public class TimerManager : MonoBehaviour
{
    public static event Action OnTimerRanOut;
    private float _timeRemaining;
    private bool _isRunning = false;

    public void StartTimer(int documentWordCount)
    {
        float currentWPM = GameManager.CurrentWPM;

        // Adaptive Scaling: Time is calculated based on the player's
        // personal speed
        if (currentWPM <= 0) { _timeRemaining = 20f; }
        else {
            _timeRemaining = ((float)documentWordCount / currentWPM)
                * 60f + _gracePeriodSeconds;
        }
        _isRunning = true;
    }

    private void Update() {
        if (!_isRunning) return;
        _timeRemaining -= Time.deltaTime;

        // Visual Warning: Flicker effect for the last 10 seconds
        if (_timeRemaining <= 10f && _timeRemaining > 0f)
            _timerText.color = (Time.time % 0.5f < 0.25f) ?
                _warningColor : _originalColor;

        if (_timeRemaining <= 0) {
            _isRunning = false;
            OnTimerRanOut?.Invoke(); // Signal the end of the
                                     transcription window
        }
    }
}
```

```
}  
  }  
}
```

C3. Drag Drop Mechanics Code

This script facilitates the tactile, physical interaction required for the drag-dropping mechanics of the game. The following is a snippet of the script DragDrop.cs with its primary functions displayed:

```
public class DragDrop : MonoBehaviour {
    private bool _isDragging = false;
    private Vector2 _dragOffset;

    private void Update() {
        if (!_isDragging) return;

        // Convert mouse position to world space and update object
        // position
        Vector2 mousePosition =
        Camera.main.ScreenToWorldPoint(Input.mousePosition);
        transform.position = new Vector3(mousePosition.x -
        _dragOffset.x, mousePosition.y - _dragOffset.y, transform.position.z);
        if (Input.GetMouseButtonUp(0)) _isDragging = false;
    }

    private void OnMouseDown(){
        // Trigger haptic-style audio feedback
        AudiopoolSFX.Instance.Play("SFX_PaperDragDrop");

        // Calculate offset to prevent the object from "snapping" to
        // center
        Vector2 mousePosition =
        Camera.main.ScreenToWorldPoint(Input.mousePosition);
        _dragOffset = mousePosition - (Vector2)transform.position;
        _isDragging = true;
    }
}
```


C4. Document Typing Transcription Mechanics Code

This script manages the core transcription mechanic of the simulation. It functions as both a visual interface for the player and a primary data collection tool for the researcher, tracking real-time performance metrics such as accuracy, typing speed, and error rates. The following is a snippet of the script TextType.cs with its primary functions displayed:

```
public class TextType : MonoBehaviour
{
    // Real-time tracking of research metrics
    private bool _timerActive = false;
    private int _wordIndex = 0;
    private string _currentInput = "";

    void Update()
    {
        if (!_canType || _allWords == null) return;

        // Cumulative data collection for performance analysis
        if (_timerActive) GameManager.TotalTypingTime +=
Time.deltaTime;

        foreach (char c in Input.inputString) {
            if (!_timerActive && !char.IsControl(c))
                _timerActive = true;

            // Word Completion Logic
            if (c == ' ' || c == '\n') {
                if (!_hasMistake && _currentInput == TargetWord)
                {
                    GameManager.TotalCharactersTyped++;
                    // Track data for WPM
                    _wordIndex++;
                    _currentInput = "";

                    if (_wordIndex >= _allWords.Length)
                        FinishDocument();
                }
            }
        }
    }
}
```

```
        // Visual/Audio feedback for errors
        else { TriggerShake(); }
    }
    else { ProcessStandardInput(c); }
}

private void ValidateInput() {
    // Ensures historical accuracy before allowing progression
    _hasMistake = !TargetWord.StartsWith(_currentInput);
    if (_hasMistake) GameManager.TotalErrors++;
}
}
```

C5. Input Mechanism Code

This script manages transition into the transcription phase of the game. When the player clicks on a dedicated text, the player will be able to transcribe a document. The following is a snippet of the script ClickToTranscribe.cs with its primary functions displayed:

```
public class ClickToTranscribe : MonoBehaviour {
    private bool _alreadyTranscribed = false;
    private void OnMouseDown() {
        if (!_alreadyTranscribed)
        {
            // Activate the global typing state for the experiment
            TypewriterKey.CanType = true;
            _alreadyTranscribed = true;

            // Global Interaction Lock: Prevents multitasking during
            data collection
            GameObject[] highlightObjects =
            GameObject.FindGameObjectsWithTag("HighlightText");
            foreach (GameObject obj in highlightObjects) {
                // Disable all colliders to focus the participant's
                attention
                foreach (BoxCollider2D col in
                obj.GetComponentsInChildren<BoxCollider2D>())
                {
                    col.enabled = false;
                }
            }
        }
    }
}
```

C6. Dialogue Controller Code

This code handles the UI display of dialogue and retrieval of dialogue contents, ensuring players can both hear and read the narrative consequences of their decisions. The following is a snippet of the script `PalReactionsController.cs` with its primary functions displayed:

```
public class PalReactionsController : MonoBehaviour {

    // Parses [MM:SS] format into seconds for audio syncing
    private void ParseSubtitles(string rawText) {
        _currentSubtitles.Clear();
        string[] lines = rawText.Split(new[] { '\r', '\n' },
            StringSplitOptions.RemoveEmptyEntries);

        foreach (string line in lines) {
            if (line.StartsWith("[") {
                // Logic to extract time and
                // text for frame-perfect display
                float timeSeconds = CalculateTime(line);
                string textStr = line.Substring(line.IndexOf(']')
                    + 1).Trim();
                _currentSubtitles.Add(new SubtitleData {
                    Time = timeSeconds, Text
                    = textStr});
            }
        }
    }

    private IEnumerator ReactionRoutine()
    {
        _audioSource.Play();
        int subIndex = 0;

        // Synchronization Loop: Matches audio playback time to
        // subtitle timestamps
        while (_audioSource.isPlaying) {
            float currentTime = _audioSource.time;
            while (subIndex < _currentSubtitles.Count && currentTime
                >= _currentSubtitles[subIndex].Time) {
```

```
        UpdateSubtitleBox(_currentSubtitles[subIndex].Text);
        subIndex++;
    }
    yield return null;
}
GenerateDocument.OnSpawnNextBatch?.Invoke();
// Trigger next gameplay beat
}
}
```

Appendix D: Game Assets

Game Assets (GitHub):

https://github.com/KimHaAnhTran/ONL_ActsOfOmission/tree/main/Assets/Art.